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3.Demonstrate the working of the decision tree based ID3 algorithm
Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

a)

Attendance	Study	Assignment	Result
High	Hard	Good	Pass
High	Hard	Poor	Pass
Low	Easy	Poor	Fail
High	Easy	Good	Pass
Low	Hard	Poor	Fail

PROGRAM

```
import pandas as pd

from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("student.csv")

X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)

y = LabelEncoder().fit_transform(df.iloc[:, -1])

model = DecisionTreeClassifier(criterion="entropy")

model.fit(X, y)

sample = [[1, 1, 1]]

print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: 1

Result: Pass

b)

Fuel	Price	Safety	Buy
Petrol	Medium	High	Yes
Diesel	High	High	Yes
Petrol	Low	Low	No
Electric	Medium	High	Yes
Diesel	Low	Low	No

PROGRAM

```
import pandas as pd

from sklearn.tree import DecisionTreeClassifier

from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("car.csv")

X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)

y = LabelEncoder().fit_transform(df.iloc[:, -1])

model = DecisionTreeClassifier(criterion="entropy")

model.fit(X, y)

sample = [[1, 1, 1]]

print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: 1

Result: Yes – Car will be purchased

c)

Income	Credit	Employment	Loan
High	Good	Permanent	Yes
High	Poor	Permanent	No
Low	Good	Temporary	No
High	Excellent	Permanent	Yes
Medium	Good	Permanent	Yes

PROGRAM

```
import pandas as pd
```

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("loan.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)
sample = [[1, 1, 1]]
print("Prediction:", model.predict(sample)[0])

```

OUTPUT

Prediction: 1

d)

Outlook	Humidity	Wind	Play
Sunny	High	Weak	No
Sunny	Normal	Weak	Yes
Rain	High	Strong	No
Cloudy	Normal	Weak	Yes
Sunny	Normal	Strong	Yes

PROGRAM

```

import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("weather.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)
sample = [[1, 1, 0]]
print("Prediction:", model.predict(sample)[0])

```

OUTPUT

Prediction: 1

Result: Play = Yes

e)

Color	Size	Sweetness	Fruit
Red	Large	High	Apple
Green	Large	High	Apple
Yellow	Small	Low	Orange
Red	Small	High	Apple
Yellow	Medium	Low	Orange

PROGRAM

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("fruit.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)
sample = [[1, 0, 1]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: 0